ModelIt!

Student Activity Pack

Can Synthetic Biology Create New Life?

Modeling Engineered Genetic Circuits and Synthetic Organism Design

Grade 11L3 | NGSS HS-LS1-1, HS-LS3-1 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What does it mean to 'engineer' a living organism, and how is it different from traditional genetic modification?**

2. **If scientists can write DNA from scratch, what do you think limits our ability to create entirely new organisms?**

3. **Draw a diagram showing how you think an engineered bacterium could be designed to perform a useful task, like cleaning up pollution.**

4. **What safety concerns would you have about releasing an engineered organism into the environment?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Circuit Complexity | Metabolic Burden | Evolutionary Stability | Output Fidelity | Growth Rate | Biocontainment Strength

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Simple Biosensor

Low complexity circuit with one sensor and one output gene, moderate containment

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Complex Metabolic Pathway

High complexity circuit with 10+ engineered genes, standard containment

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Containment Breach Test

Complex circuit with reduced biocontainment strength

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Genetic Part Standardization — The degree to which genetic components are characterized and modular — standardized parts with predictable behavior simplify circuit design but biological context-dependence means parts often behave differently in new organisms

Host Organism Compatibility — How well the synthetic circuit integrates with the host cell's existing metabolism — some organisms are better chassis for engineering due to their genetic tractability, growth characteristics, and existing genetic tools

Regulatory Approval Pathway — The government review process for deploying synthetic organisms — EPA, FDA, and USDA each have jurisdiction depending on the application, and the approval timeline can exceed the engineering timeline by years

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If we can write DNA from scratch like computer code, can we engineer entirely new forms of life — and what are the consequences if we do? Use evidence from your simulation to support your answers.

1 How does your model demonstrate why evolution is both the central tool and the central obstacle in synthetic biology?

2 What does the relationship between circuit complexity and evolutionary stability tell you about the practical limits of genetic engineering?

3 Why must biocontainment use multiple independent mechanisms rather than a single safety system?

4 How would your model change if you could engineer organisms that are immune to mutation in their synthetic circuits?

5 What are the ethical boundaries of creating new forms of life, and who should draw those boundaries?

8 STEM Challenge: Design a Synthetic Organism for Environmental Remediation

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YOUR ENGINEERING MISSION

Design a synthetic organism engineered to break down a specific environmental pollutant, incorporating multiple biocontainment mechanisms and addressing evolutionary stability over deployment timescales.

THE SCENARIO

A chemical spill has contaminated a river with persistent organic pollutants that natural bacteria cannot efficiently degrade. Your team must design a synthetic bacterium engineered with a metabolic pathway to break down the pollutant, biocontainment mechanisms to prevent environmental persistence after the cleanup, and strategies to maintain the engineered function long enough to complete remediation before evolution disables it.

Engineering Design Thinking

1. WHAT SPECIFIC GENETIC CIRCUIT DESIGN WOULD YOU USE TO CONNECT POLLUTANT SENSING TO DEGRADATION ENZYME EXPRESSION?

2. HOW MANY INDEPENDENT BIOCONTAINMENT MECHANISMS WOULD YOU INCLUDE AND WHAT TYPE (KILL SWITCH, AUXOTROPHY, SYNTHETIC AMINO ACID)?

3. WHAT IS YOUR STRATEGY FOR MAINTAINING ENGINEERED FUNCTION AGAINST EVOLUTIONARY PRESSURE DURING THE DEPLOYMENT PERIOD?

4. HOW WOULD YOU MONITOR THE SYNTHETIC ORGANISM IN THE FIELD TO DETECT CONTAINMENT FAILURES OR FUNCTION LOSS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Synthetic Biologists design and build new biological systems at companies like Ginkgo Bioworks, Zymergen, Amyris, and Twist Bioscience, earning $95,000-$175,000/year. Biosafety Officers ensure containment protocols protect researchers and the environment at biotech companies and government agencies like the CDC, earning $80,000-$140,000/year. Bioinformatics Scientists who design synthetic genetic sequences using computational tools earn $90,000-$165,000/year.